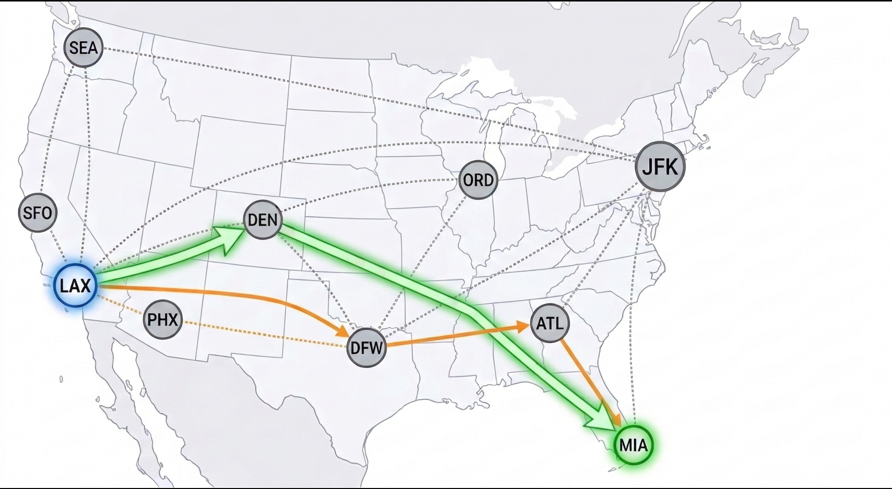
CS-GY 6033

Design and Analysis of Algorithms I

Lecture 7 | Part 1

Shortest Paths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Recall

- The **length** of a path is

(# of nodes) – 1

Definitions

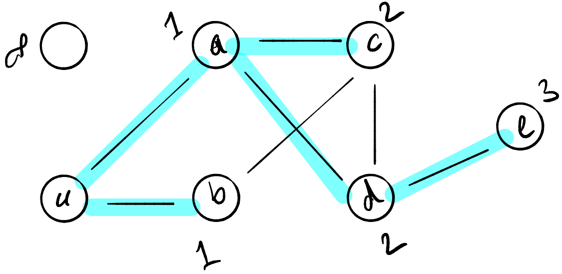
- A **shortest path** between u and v is a path between u and v with smallest possible length.
  - There may be several, or none at all.

- The **shortest path distance** is the length of a shortest path.
  - Convention: ∞ if no path exists.
  - “the distance between u and v” means spd.

Today: Shortest Paths

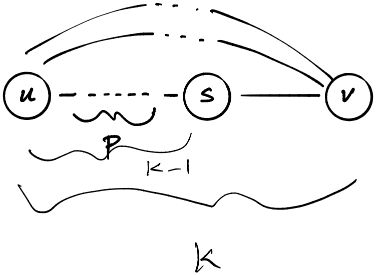
- **Given**: directed/undirected graph G, source u
- **Goal**: find shortest path from u to every other node

Example



Key Property

- A shortest path of length k is composed of:
  - A **shortest path** of length $k - 1$.
  - Plus one edge.



Algorithm Idea

- Find all nodes distance 1 from source.
- Use these to find all nodes distance 2 from source.
- Use these to find all nodes distance 3 from source.
- ...

It turns out…

…this is exactly what BFS does.

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Lecture 7 | Part 2

BFS for Shortest Paths

Key Property of BFS

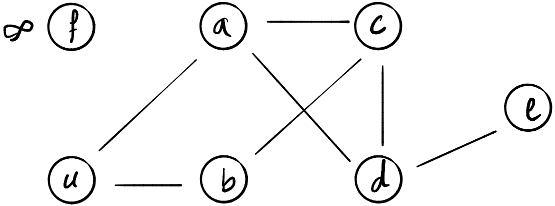
- For any $k \geq 1$ you choose:
- All nodes distance $k - 1$ from source are added to the queue before any node of distance k.
- Therefore, nodes are "processed" (popped from queue) in order of distance from source.

Example

P : [u]

P : [a, b] dist 1

P : $[\underbrace{b}_{\text{dist 1}}, \underbrace{c, d}_{\text{dist 2}}]$



Discovering Shortest Paths

- We “discover” shortest paths when we pop a node from queue and look at its neighbors.
- But the neighbor’s status matters!

Consider This

- We pop a node s.
- It has a neighbor v whose status is **undiscovered**.
- We've discovered a **shortest path** to v through s!

Consider This

- We pop a node s.
- It has a neighbor v whose status is **pending** or **visited**.
- We already have a shortest path to v.

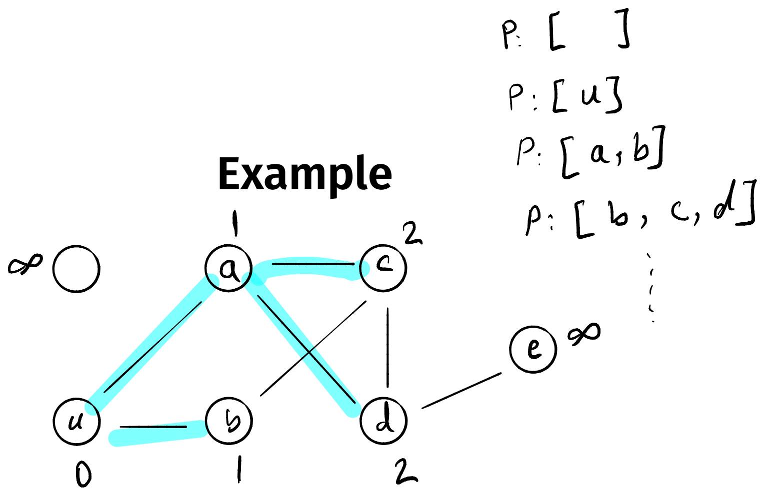
Modifying BFS

- Use BFS "framework".
- Return dictionary of **search predecessors**.
  - If v is discovered while visiting u, we say that u is the **BFS predecessor** of v.
  - This encodes the shortest paths.
- Also return dictionary of shortest path distances.

```python
def bfs_shortest_paths(graph, source):
    """Start a BFS at `source`."""
    status = {node: 'undiscovered' for node in graph.nodes}
    distance = {node: float('inf') for node in graph.nodes}
    predecessor = {node: None for node in graph.nodes}

    status[source] = 'pending'
    distance[source] = 0
    pending = deque([source])

    # while there are still pending nodes
    while pending:
        u = pending.popleft()
        for v in graph.neighbors(u):
            # explore edge (u,v)
            if status[v] == 'undiscovered':
                status[v] = 'pending'
                distance[v] = distance[u] + 1
                predecessor[v] = u
                # append to right
                pending.append(v)
        status[u] = 'visited'

    return predecessor, distance
```

dis[v] = dis[u] +1

Example



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BFS Trees

Result of BFS

- Each node reachable from source has a single BFS predecessor.
  - Except for the source itself.

- The result is a **tree** (or forest).

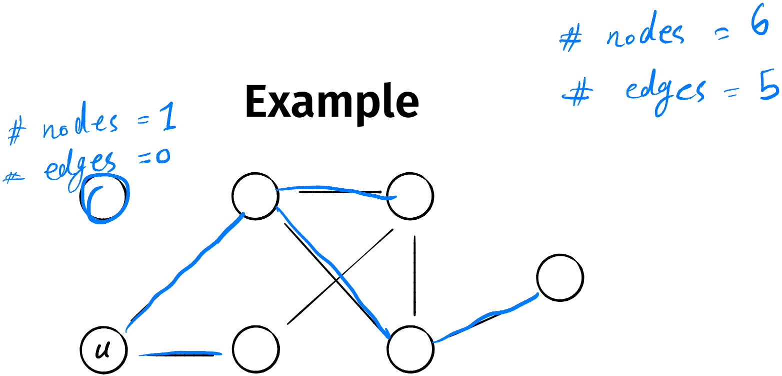
Trees

- A ~~(free)~~ **tree** is an undirected graph $T = (V, E)$ such that T is connected and $|E| = |V| - 1$.
- A **forest** is graph in which each connected component is a tree.

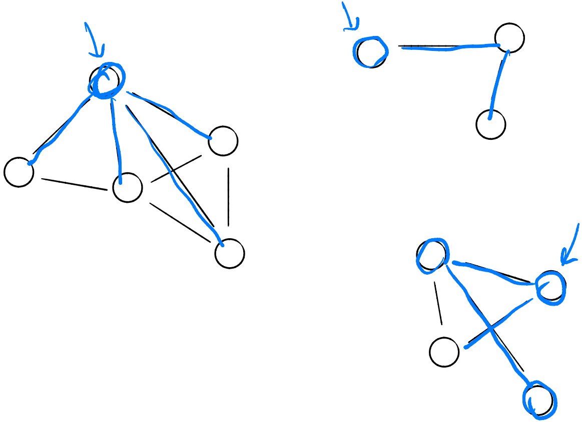
BFS Trees (Forests)

- If the input is connected, BFS produces a **tree**.
- If the input is not connected, BFS produces a **forest**.

Example



Example



BFS Trees

- BFS trees and forests encode shortest path distances.

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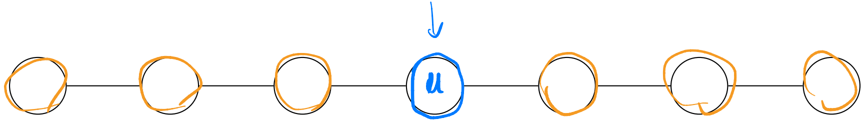
Lecture 7 | Part 4

Depth First Search

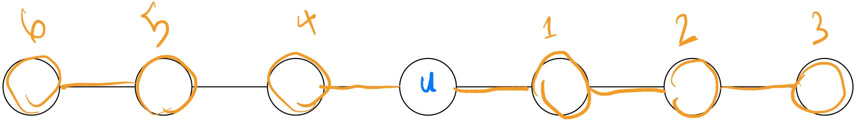
Visiting the Next Node

- Which node do we process next in a search?
- BFS: the **oldest** pending node.
- DFS (today): the **newest** pending node.
  - Naturally recursive.
  - Useful for solving different problems.

Example (BFS)



Example (DFS)

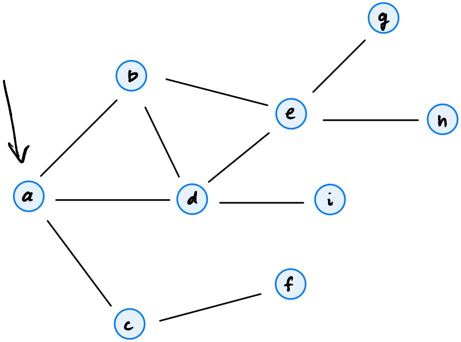


```python
def dfs(graph, u, status=None):
    """Start a DFS at `u`."""
    # initialize status if it was not passed
    if status is None:
        status = {node: 'undiscovered' for node in graph.nodes}

    status[u] = 'pending'
    for v in graph.neighbors(u): # explore edge (u, v)
        if status[v] == 'undiscovered':
            dfs(graph, v, status)
    status[u] = 'visited'
```

Exercise

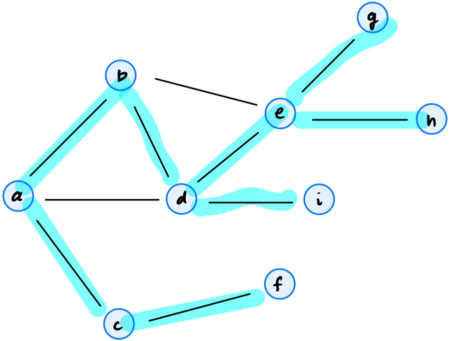
Write the nested function calls for a DFS on the graph below.




```python
def dfs(graph, u, status=None):
    """Start a DFS at `u`."""
    ...
    status[u] = 'pending'
    for v in graph.neighbors(u): # explore edge (u, v)
        if status[v] == 'undiscovered':
            dfs(graph, v, status)
    status[u] = 'visited'
```

Exercise

Write the nested function calls for a DFS on the graph below.



DFS(a):

  DFS(b):

    DFS(d):

      DFS(e):

        DFS(g)

        DFS(h)

      DFS(i)

  DFS(c):

    DFS(f)

Differences

- In **BFS**, we "finish" a node u before moving on to the next.
- In **DFS**, we go to many other nodes, but "come back" to u.

Main Idea

We'll see that the nested structure of the **recursive function calls** gives us useful new information about the graph's structure.

Full DFS

- `dfs(u)` will visit all nodes **reachable** from u.
  - But not all nodes may be reachable from u!

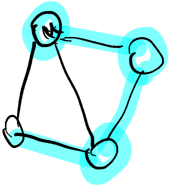
- To visit **all** nodes in graph, need **full DFS**.

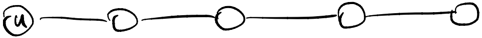
```python
def full_dfs(graph):
    status = {node: 'undiscovered' for node in graph.nodes}
    for node in graph.nodes:
        if status[node] == 'undiscovered'
            dfs(graph, node, status)
```

```python
def full_dfs(graph):
    status = {node: 'undiscovered' for node in graph.nodes}
    for node in graph.nodes:
        if status[node] == 'undiscovered'
            dfs(graph, node, status)


def dfs(graph, u, status=None):
    """Start a DFS at `u`."""
    # initialize status if it was not passed
    if status is None:
        status = {node: 'undiscovered' for node in graph.nodes}

    status[u] = 'pending'
    for v in graph.neighbors(u): # explore edge (u, v)
        if status[v] == 'undiscovered':
            dfs(graph, v, status)
    status[u] = 'visited'
```





Time Complexity

- In a full DFS:
  - `dfs` called on each node exactly once.
  - Like BFS, each edge is explored exactly:
    - once if directed
    - twice if undirected

- Time: $\Theta(V + E)$, **just like BFS**.

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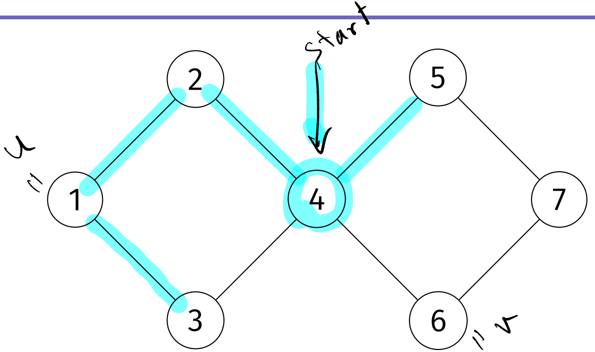
Lecture 7 | Part 5

Nesting Properties of DFS

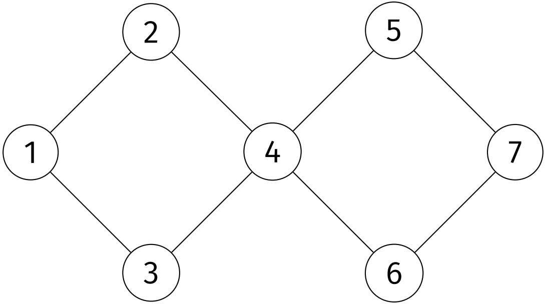
Exercise

True or **False**: if v is reachable from u and v is **undiscovered** when `dfs(u)` is called, then `dfs(v)` must be called during `dfs(u)`.



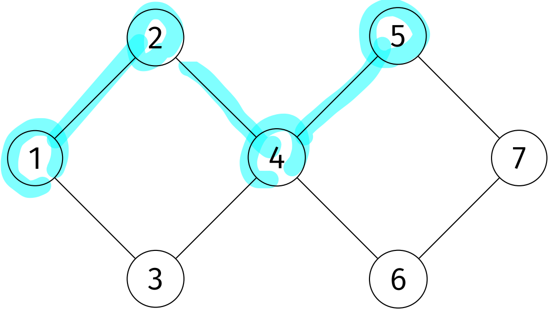
False!

- Suppose `dfs(4)` is the root call.
  - When `dfs(1)` is called, node 5 is undiscovered.
  - But `dfs(5)` is **not** called during `dfs(1)`.



However..

- This intuition is correct if there is a path of **undiscovered** nodes from u to v when `dfs(u)` is called.



Key Property of DFS (Informal)

- If at the time `dfs(u)` is called...
  1. v is **undiscovered**; and
  2. there is a path of **undiscovered** nodes from u to v,

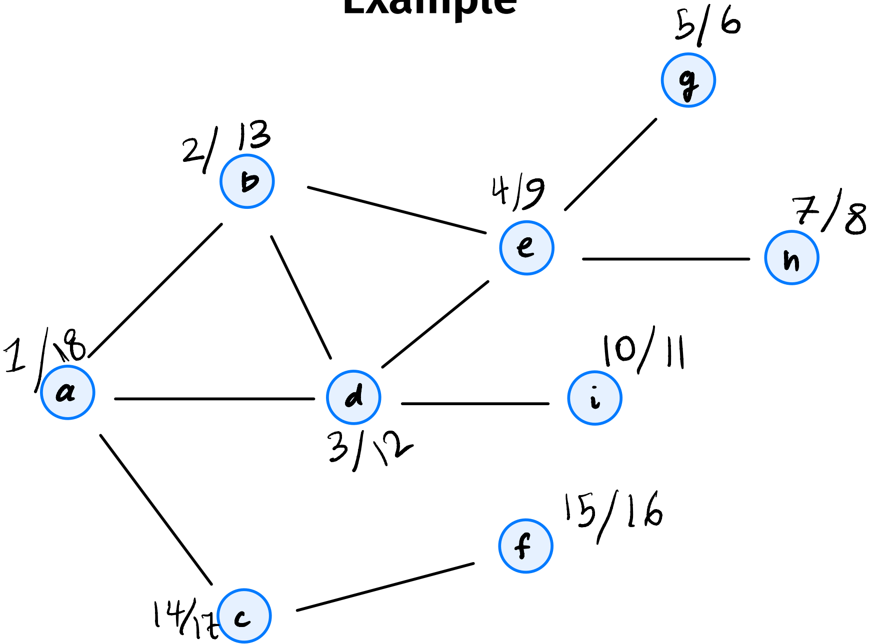
- ...then `dfs(v)` will **start and finish** during the call to `dfs(u)`.

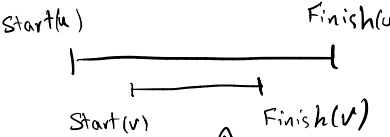
Start and Finish Times

- Keep a running clock (an integer).
- For each node, record
  - **Start time**: time when marked **pending**
  - **Finish time**: time when marked **visited**
- Increment clock whenever node is marked **pending**/**visited**

```python
from dataclasses import dataclass

@dataclass
class Times:
    clock: int
    start: dict
    finish: dict

def full_dfs_times(graph):
    status = {node: 'undiscovered' for node in graph.nodes}
    predecessor = {node: None for node in graph.nodes}
    times = Times(clock=0, start={}, finish={})
    for u in graph.nodes:
        if status[u] == 'undiscovered':
            dfs_times(graph, u, status, times)
    return times, predecessor

def dfs_times(graph, u, status, predecessor, times):
    times.clock += 1
    times.start[u] = times.clock
    status[u] = 'pending'
    for v in graph.neighbors(u): # explore edge (u, v)
        if status[v] == 'undiscovered':
            predecessor[v] = u
            dfs_times(graph, v, status, times)
    status[u] = 'visited'
    times.clock += 1
    times.finish[u] = times.clock
```

Example





Key Property of DFS

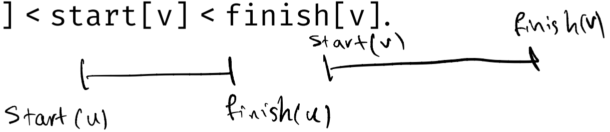
- Suppose `dfs(u)` is called before `dfs(v)`.

- If when `dfs(u)` is called there is a path of **undiscovered** nodes from *u* to *v*, then:

  `start[u] < start[v] < finish[v] < finish[u].`

- Otherwise:

  `start[u] < finish[u] < start[v] < finish[v].`



Key Property

- Take any two nodes *u* and *v* ($u \neq v$).
- Assume for simplicity that `start[u]` < `start[v]`.
- Exactly one of these is true:
  - `start[u]` < `start[v]` < `finish[v]` < `finish[u]`
  - `start[u]` < `finish[u]` < `start[v]` < `finish[v]`

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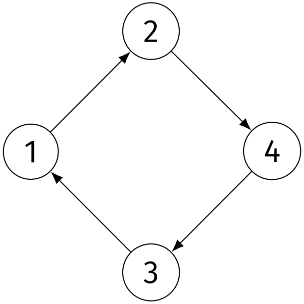
Cycles

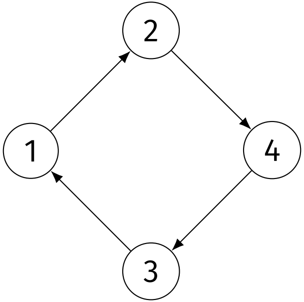
Cycle

- A **cycle** in a directed graph is a path that starts and ends at the same node.



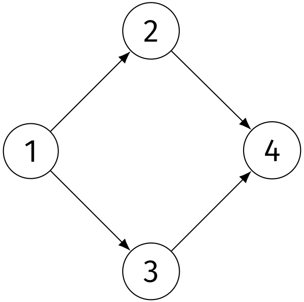
Cycle

- Alternatively: there is a **cycle** if u is reachable from v and v is reachable from u, for some $u \neq v$.



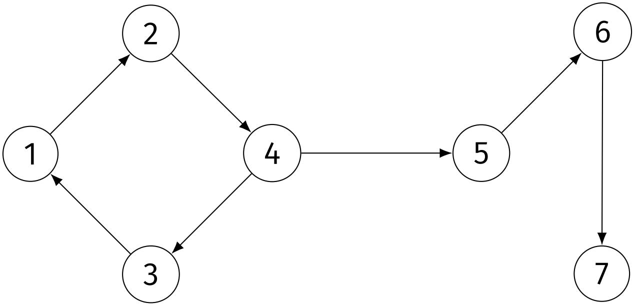
DAG

- A **directed acyclic graph** (DAG) is a directed graph with **no cycles**.



Cyclic Graphs

- A graph is cyclic even if it has only one cycle.
  - It doesn't have to be the whole graph.

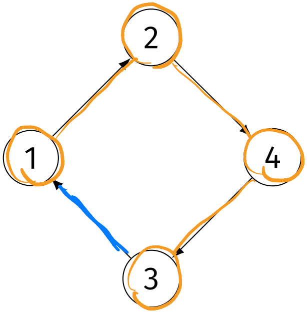


Detecting Cycles

- We check for cycles by looking for **back edges** in a full DFS.
- (u, v) is a **back edge** if while visiting node u, we see that v is **pending**.

```
...
for v in graph.neighbors(u): # explore edge (u, v)
    if status[v] == 'undiscovered':
        dfs(graph, v, status)
    elif status[v] == 'pending':
        # back edge (u, v) found!
...
```

Example



Theorem

A directed graph has a cycle ***if (and only if)*** *a full DFS finds a back edge.*

Why?

- If a back edge (u, v) is found, then a cycle exists.
  - Suppose v is pending when we visit u.
  - This means that there is a path from v to u.
  - There is also a path from u to v.
  - So there is a cycle.

Why?

- If a cycle exists, then there is a back edge.
  - Suppose there is a cycle $v_1 \rightarrow v_2 \rightarrow \cdots \rightarrow v_k \rightarrow v_1$.
  - Without loss of generality, assume v_1 is the first node in the cycle that is visited by the full DFS.
  - At the moment of `dfs(v_1)`, there is a path of undiscovered nodes between v_1 and v_k.
  - Therefore `dfs(v_k)` will be called during `dfs(v_1)`.
  - During `dfs(v_k)`, we'll see the back edge.

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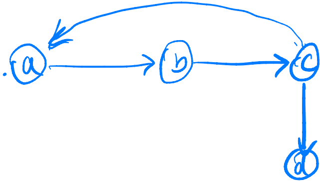
Lecture 7 | Part 7

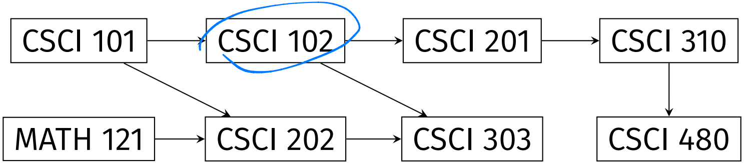
Topological Sort

Applications of DFS



- Is node v reachable from node u? **DFS**, **BFS**
- Is the graph connected? **DFS**, **BFS**
- How many connected components? **DFS**, **BFS**
- Find the shortest path between u and v. **DFS**, **BFS**
- Does the graph have a cycle? **DFS**, **BFS**

Prerequisite Graphs



Goal: find order in which classes should be taken in order to satisfy the prerequisites of CSCI 480.

Note

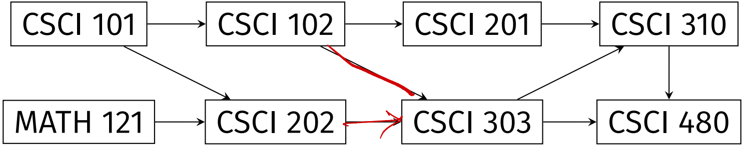
- Prerequisite graphs are[1] DAGs.

[1] Or they should be, at least!

Topological Sorts

- **Given**: a DAG, $G = (V, E)$.
- **Compute**: an ordering of V such that if $(u, v) \in E$, then u comes before v in the ordering
- This is called a **topological sort** of G.

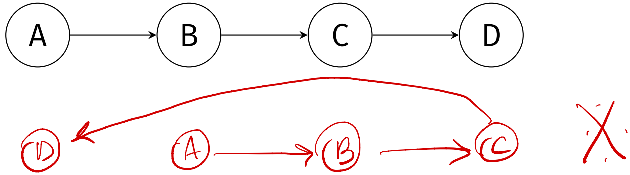
Example



✓ MATH 121, CSCI 101, CSCI 202, CSCI 102, CSCI 303, CSCI 201, CSCI 310, CSCI 480

Computing a Topological Sort

- How do we compute a topological sort, algorithmically?
- **Observation**: if v is reachable from u, v **must** come **after** u in the topological sort.



Exercise

Suppose v is reachable from u in a DAG.

True or false: after a full DFS, `finish[v]` < `finish[u]`.

Claim

- If v is reachable from u in a DAG, then:

$$\texttt{finish[v]} < \texttt{finish[u]}$$

Idea

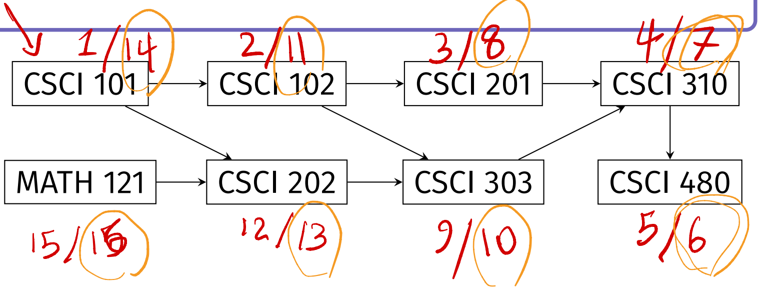
- Take any two nodes u and v ($u \neq v$).
- Assume the graph is a DAG, run DFS.
- If v is reachable from u, then `finish[v]` < `finish[u]`.

Putting it together...

- **Observation:** If v is reachable from u, then v must come after u in the topological sort.
- **Recall:** If v is reachable from u, then `finish[v] < finish[u]`.

Exercise

Compute start and finish times using CSCI 101 as the source.



Idea

- **Observation:** If v is reachable from u, then v must come after u in the topological sort.
- **Recall:** If v is reachable from u, then `finish[v] < finish[u]`.
- **Therefore:** if `finish[v] < finish[u]`, then v must come after u in the topological sort.
- **Idea:** sort nodes in **descending** order by finish time.

Algorithm

- To find a topological sort (if it exists):
  - Compute times with Full DFS. $\Theta(|V|+|E|)$
  - Sort in **descending** order by finish time.

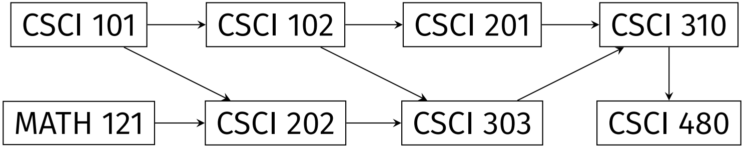
$\Theta(|V|)$

$E \log E$

- Time complexity:

$\Theta(|V|+|E|)$

Example



CSCI 101
CSCI 102
CSCI 201
CSCI 310
MATH 121
CSCI 202
CSCI 303
CSCI 480

Note

- There can be many valid topological sorts!